A. Power Consumption Calculation

time limit per test: 1 second
memory limit per test: 256 megabytes

Tom is interested in power consumption of his favourite laptop. His laptop has three modes. In normal mode laptop consumes P_1 watt per minute. T_1 minutes after Tom moved the mouse or touched the keyboard for the last time, a screensaver starts and power consumption changes to P_2 watt per minute. Finally, after T_2 minutes from the start of the screensaver, laptop switches to the "sleep" mode and consumes P_3 watt per minute. If Tom moves the mouse or touches the keyboard when the laptop is in the second or in the third mode, it switches to the first (normal) mode. Tom's work with the laptop can be divided into n time periods $[l_1, r_1], [l_2, r_2], \ldots, [l_n, r_n]$. During each interval Tom continuously moves the mouse and presses buttons on the keyboard. Between the periods Tom stays away from the laptop. Find out the total amount of power consumed by the laptop during the period $[l_1, r_n]$.

Input

The first line contains 6 integer numbers $n, P_1, P_2, P_3, T_1, T_2$ ($1 \le n \le 100, 0 \le P_1, P_2, P_3 \le 100, 1 \le T_1, T_2 \le 60$). The following n lines contain description of Tom's work. Each i-th of these lines contains two space-separated integers l_i and r_i ($0 \le l_i < r_i \le 1440$, $r_i < l_{i+1}$ for $i < n$), which stand for the start and the end of the i-th period of work.

Output

Output the answer to the problem.

Examples

input Copy

```
1 3 2 1 5 10
0 10
```

output Copy

```
30
```

input Copy

```
2 8 4 2 5 10
20 30
50 100
```

output Copy

```
570
```

→ Attention

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

Codeforces Beta Round 10

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
369435195	Apr/03/2026 11:09	Accepted

→ Problem tags

implementation *900

No tag edit access

→ **Contest materials**

- Announcement (en)

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